

Probability

Important formula:

1. Under discrete random variable,

↓
Probability mass function

$$f_X(t) = P(X^{-1}(t)) \\ = P(\omega \mid X(\omega) \leq t)$$

Cumulative Distribution function:

$$F_X(t) = \sum_{u \leq t} f(u)$$

Probability mass function for different distributions

1) $\text{Ber}(p)$,

$$f(t) = \begin{cases} q & , \text{ if } t = 0 \\ p & , \text{ if } t = 1 \quad (\text{success}) \\ 0 & , \text{ otherwise} \end{cases}$$

$$\& F(t) = \begin{cases} 0 & , t < 0 \\ q & , 0 \leq t < 1 \\ p + q = 1 & , 1 \leq t \end{cases}$$

2) $\text{Bin}(n, p)$,

$$f(t) = \begin{cases} \binom{n}{k} p^k \cdot q^{n-k} & , \text{ if } t = k, 0 \leq k \leq n \\ 0 & , \text{ otherwise} \end{cases}$$

3) $\text{Geo}(p)$, Let $p \in [0, 1]$. For $k \in \mathbb{N}$

$$f(t) = \begin{cases} q^{k-1} p & , \text{ if } t = k \\ 0 & , \text{ otherwise} \end{cases}$$

4) Poiss(λ)

$$f(t) = \begin{cases} \frac{e^{-\lambda} \cdot \lambda^k}{k!}, & \text{if } t = k \in \mathbb{N} \cup \{0\} \text{ \& } \\ 0, & \text{otherwise} \end{cases}$$

5) Hypergeo(b, w, m) with $m \leq b+w$
and $b, w, m \in \mathbb{N}$

$$f(t) = \begin{cases} \frac{\binom{b}{k} \binom{w}{m-k}}{\binom{b+w}{m}}, & \text{if } t = k \in \mathbb{N} \cup \{0\} \\ 0, & \text{otherwise} \end{cases}$$

Expectation of discrete Random Variable

$$E[X] = \sum_{n=1}^{\infty} t_n f_X(t_n), \quad E[h(X)] = \sum_{n=1}^{\infty} h(t_n) f_X(t_n)$$

Probability distribution functions of cont^s R.V. :

$$F(t) = \int_{-\infty}^t f(u) du$$

± Uniform distribution $U(a, b)$ on an interval $[a, b]$

$$f(t) = \begin{cases} \frac{1}{b-a}, & \text{if } t \in [a, b] \\ 0, & \text{otherwise} \end{cases}$$

2. Exponential distribution ($\text{Exp}(\lambda)$)

$$f(t) = \begin{cases} \lambda e^{-\lambda t} & , \text{ if } t > 0 \\ 0 & , \text{ if } t \leq 0 \end{cases}$$

3. Normal distribution, ($N(\mu, \sigma^2)$), with $\mu \in \mathbb{R}$ and $\sigma^2 > 0$.

$$\text{PDF: } f(t) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\left(\frac{t-\mu}{\sqrt{2}\sigma}\right)^2}$$

4. The Gamma distribution, ($\text{Gamma}(\nu, \lambda)$) with shape parameter ν and scalar parameter λ .

$$f(t) = \begin{cases} \frac{\lambda^\nu t^{\nu-1} \cdot e^{-\lambda t}}{\Gamma(\nu)} & , \text{ if } t > 0 \text{ and} \\ 0 & , \text{ if } t \leq 0 \end{cases}$$

5. The Beta distribution $B(a, b)$, $a, b > 0$

$$f(t) = \begin{cases} 0 & , \text{ if } t \notin (0, 1) \\ \frac{t^{a-1} (1-t)^{b-1}}{B(a, b)} & , \text{ if } t \in (0, 1) \end{cases}$$

6. Cauchy Distribution ($\text{Cauchy}(\lambda, a)$) with $\lambda > 0$ and $a \in \mathbb{R}$.

$$f(t) = \frac{\lambda}{\pi(\lambda^2 + (t-a)^2)}$$

- What is an algebra?

Let Ω be a set and let $\mathcal{F} \subset \mathcal{P}(\Omega)$ be the collection of subsets of Ω . Then, $\mathcal{F}$ is an algebra on Ω if it satisfies the following properties:

(a) $\emptyset \in \mathcal{F}, \Omega \in \mathcal{F}$

(b) If $A \in \mathcal{F}$ then $A^c \in \mathcal{F}$

(c) If $A, B \in \mathcal{F}$ then $A \cup B \in \mathcal{F}$

(d) If $A, B \in \mathcal{F}$ then $A \cap B \in \mathcal{F}$

- What is a σ -algebra?

An algebra $\mathcal{M}$ is called a σ -algebra if it has the following property:

For every countable collection of pairwise disjoint sets $\{A_n\}_{n=1}^{\infty}$ in $\mathcal{M}$,

$$A = \bigcup_{n=1}^{\infty} A_n \in \mathcal{M}$$

- The pair $(\Omega, \mathcal{M})$ is called a measurable space and the elements of $\mathcal{M}$ are called measurable sets.

Important formulae and concepts for integration.

- Let $(\Omega, \mathcal{M}, \mu)$ be a measure space.

If $s(x) = \sum_{j=1}^n c_j \mathbb{1}_{A_j}(x)$ be a simple function which takes non-negative values. We define

$$\int_E s d\mu = \int_E \sum_{j=1}^n c_j \mathbb{1}_{A_j} d\mu = \sum_{j=1}^n c_j \mu(E \cap A_j)$$

when $E = \Omega$ then, $\int_{\Omega} s d\mu = \sum_{j=1}^n c_j \mu(A_j)$, where A_j are measurable sets.

- If $f: \Omega \rightarrow [0, \infty]$ be a measurable function then

we define,

$$\int_E f d\mu = \sup_{0 \leq s \leq f} \int_E s d\mu$$

Called as
Lebesgue integral
of f over
measurable set E
w.r.t. measure
 μ .

Special Case: When f is a bounded measurable function and s_n is a sequence such that $0 \leq s_0(x) \leq s_1(x) \leq s_2(x) \leq \dots \leq s_n(x) \leq \dots \leq f(x)$ and $\lim_{n \rightarrow \infty} s_n(x) = f(x)$ $\forall x \in \Omega$ then we know that $s_n \rightarrow f$ uniformly on Ω .
it follows that $\int_{\Omega} s_n d\mu = a_n$ is a Cauchy seq.
of real numbers.

and in this case $\int_{\Omega} f d\mu = a$

Note: A measurable function $f: \Omega \rightarrow \mathbb{C}$ is said to be in $L^1(\Omega, \mu)$ if $\int_{\Omega} |f| d\mu < \infty$,

Such functions are called as Lebesgue integrable functions. (also called absolutely integrable functions).

Lebesgue Monotone Convergence Theorem (MCT)

Let $f_n: \Omega \rightarrow [0, \infty]$ be a sequence of measurable function such that

$$(1) \quad 0 \leq f_1(x) \leq f_2(x) \leq \dots \leq f_n(x) \leq \dots$$

$\forall n \in \mathbb{N} \text{ and } x \in \Omega$

$$(2) \quad \lim_{n \rightarrow \infty} f_n(x) \text{ exists and is equal to the}$$

$$\text{limit } f(x) \quad \forall x \in \Omega$$

Then, f is measurable and,

$$\lim_{n \rightarrow \infty} \int_{\Omega} f_n d\mu = \int_{\Omega} f d\mu$$

Lebesgue's Dominated Convergence Theorem

Let $f_n: \Omega \rightarrow \mathbb{C}$ be a sequence of measurable functions such that $\exists$ a $g(x) \in L^1(\Omega, \mu)$ so that $|f_n(x)| \leq g(x) \quad \forall n \in \mathbb{N} \text{ and } x \in \Omega$

and $\lim_{n \rightarrow \infty} f_n(x) = f(x)$ exists $\forall x \in \Omega$

Then, $f \in L^1(\Omega, \mu)$ and,

$$\lim_{n \rightarrow \infty} \int_{\Omega} f_n d\mu = \int_{\Omega} f d\mu$$

Continuous Random Variable

The random variable $X: \Omega \rightarrow \mathbb{R}$ is said to be of continuous type with CDF F_X if

F_X is absolutely continuous i.e. $\exists$ a non-negative function $f(x)$ such that

$$F(x) = \int_{-\infty}^x f(t) dt$$

Case I: Let $g'(x) > 0$,

$$\text{Now, } F_Y(y) = P(Y \leq y)$$

Substitute $Y = g(X)$, we get

$$F_Y(y) = P(g(X) \leq y)$$

$\therefore g$ is strictly increasing so $g(X) \leq y$

$\Rightarrow X \leq g^{-1}(y)$ (inequality doesn't change)

$$\therefore F_Y(y) = P(X \leq g^{-1}(y))$$

$$= \int_{-\infty}^{g^{-1}(y)} f_X(t) dt$$

Let $g(t) = u$

$$\left. \begin{array}{l} t \\ -\infty \end{array} \right| \begin{array}{l} u \\ g(-\infty) = a \\ y \end{array}$$

$$t = g^{-1}(u)$$

$$\therefore F_Y(y) = \int_{u=a}^{u=y} f_X(g^{-1}(u)) | [g^{-1}]'(u) | \cdot du$$

Since, $u < a$ and $|[g^{-1}]'(u)| = +[g^{-1}]'(u)$

$$\therefore f(u) = 0$$

$$\therefore F_Y(y) = \int_{-\infty}^y \underbrace{f_X(g^{-1}(u)) | [g^{-1}]'(u) |}_{f_Y(u)} du$$

Case II: If $g'(x) < 0$ then

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y)$$

$$= P(X \geq g^{-1}(y))$$

$$= 1 - P(X < g^{-1}(y))$$

$$= 1 - P(X \leq g^{-1}(y))$$

$$= 1 - \int_{-\infty}^{g^{-1}(y)} f_X(t) dt$$

$$= \int_{-\infty}^{\infty} f_X(t) dt - \int_{-\infty}^{g^{-1}(y)} f_X(t) dt$$

$$= \int_{g^{-1}(y)}^{\infty} f_X(t) dt =$$

$\because X$ is a cont^s random variable

$$P(X = z) = 0$$

$$\text{so, } P(X \leq z) = P(X < z)$$

$$\text{Let } g(t) = u$$

$$t = g^{-1}(u)$$

Let $(\Omega, \mathcal{M}, P)$ be a probability measure space.

and let $X: \Omega \rightarrow \mathbb{R}$ be a random variable with $X \in L^1(\Omega, P)$.

The expectation

$$\mu = \mathbb{E}[X] = \int_{\Omega} X dP$$

Also, we know that if $X \in L^1(\Omega, P)$ then

$$\mathbb{E}[X] = \int_{\Omega} X(\omega) dP(\omega)$$

The push-forward measure μ on $(\mathbb{R}, \mathcal{B})$ defined by

$$\mu(A) = P(X^{-1}(A))$$

$$= \mu(A) = P(\{\omega \in \Omega : X(\omega) \in A\})$$

for any $A \subseteq \mathbb{R}$

A key property of push-forward measure μ is

$$\int_{\Omega} g(X(\omega)) dP(\omega) = \int_{\mathbb{R}} g(x) d\mu(x)$$

for every measurable $g: \mathbb{R} \rightarrow \mathbb{R}$.

Take $g(x) = x$, we get $\mathbb{E}[X] = \int_{\Omega} X(\omega) dP(\omega)$

$$= \int_{\mathbb{R}} x d\mu(x)$$

If X is of continuous type then μ is absolutely continuous wrt. measure m on $\mathbb{R}$

$\therefore \exists$ a non-negative function $f_X(x)$ such that

$$d\mu(x) = f_X(x) dm(x)$$

$$\begin{aligned} \therefore E[X] &= \int_{\mathbb{R}} x \cdot f_X(x) \cdot dm(x) \\ &= \int_{-\infty}^{\infty} x f_X(x) dx \end{aligned}$$

$$\begin{aligned} \therefore E[X] &= \begin{cases} \int_{\Omega} X dP \\ = \int_{\Omega} x d\mu(x) & (\text{By pushforward measure}) \\ = \int_{\mathbb{R}} x f_X(x) \cdot dx & (\text{if } \mu \text{ has density } f_X) \end{cases} \\ &= \int_{\mathbb{R}} x dF_X(x) \end{aligned}$$

Now, $E[Y] = \int_{\mathbb{R}} g(x) f_X(x) dx$

$$\begin{array}{c|c} x & y \\ \hline -\infty & a \\ \infty & b \end{array}$$

Proof:

We know that

Let $g(x) = y$

$$\begin{aligned} \int_{-\infty}^{\infty} g(x) f_X(x) dx &= \int_a^b y \cdot \underbrace{f_X(g^{-1}(y)) (g^{-1}(y))'}_{f_Y(y)} dy \\ &= \int_a^b y \cdot f_Y(y) dy = E[Y] \end{aligned}$$

Theorem 1.14.4

Let $X: \Omega \rightarrow \mathbb{R}$ be a random variable.
If $E[|X|^t]$ exists then so do $E[|X|^s]$
for $0 < s < t$.

For $|X| \leq 1$ then,

$$|X|^s \leq 1$$

For $|X| > 1$, then $|X|^s \leq |X|^t$ ($\because s < t$ & $|X| > 1$)

$$\begin{aligned} \therefore E[|X|^s] &= \int_{\Omega} |X|^s dP \\ &= \int_{|X| \leq 1} |X|^s dP + \int_{|X| > 1} |X|^s dP \end{aligned}$$

$$\leq \int_{|X| \leq 1} 1 \cdot dP + \int_{|X| > 1} |X|^t dP$$

$$= P(|X| \leq 1) + \int_{|X| > 1} |X|^t \cdot dP$$

$$\leq 1 + E[|X|^t] < \infty$$

$$(\because E[|X|^t] < \infty)$$

$$\Rightarrow E[|X|^s] < \infty$$

$$\text{i.e. } E[|X|^s] \text{ exists.}$$

Holder's Inequality:

Let $(\Omega, \mathcal{M}, m)$ be a measurable space. For all $f, g: \Omega \rightarrow \mathbb{C}$ measurable functions:

$$\left(\int_{\Omega} |fg| dm \right) \leq \left(\int_{\Omega} |f|^p dm \right)^{1/p} \left(\int_{\Omega} |g|^q dm \right)^{1/q}$$

$$\text{or } \int_{\Omega} |fg| dm \leq \|f\|_p \cdot \|g\|_q$$

Minkowski's Inequality If $f, g \in L^p(\Omega, m)$,

$p \geq 1$ then,

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p$$

Lyapunov's Inequality: If $1 \leq s \leq t$ and $f \in L^t(\Omega, P)$ then

$$\|f\|_s \leq \|f\|_t$$

Marginal PDFs

fix it $\leftarrow$ $f_{X_1}(x_1) = \int_{-\infty}^{\infty} f(x_1, x_2) dx_2$

fix it $\leftarrow$ $f_{X_2}(x_2) = \int_{-\infty}^{\infty} f(x_1, x_2) dx_1$

Marginal Distribution functions

$$F_{X_1}(x_1) = \lim_{x_2 \rightarrow \infty} F(x_1, x_2) = \int_{-\infty}^{x_1} f_1(t) dt$$

and $F_{X_2}(x_2) = \lim_{x_1 \rightarrow \infty} F(x_1, x_2) = \int_{-\infty}^{x_2} f_2(t) dt$

Conditional Distributions

For cont^s R.V. $P(X_2 = x_2) = 0$ so we can't directly use.

Can't use $\rightarrow$
$$P(X_1 \leq x_1 | X_2 = x_2) = \frac{P(X_1 \leq x_1, X_2 = x_2)}{P(X_2 = x_2)}$$

We define it in small interval x_2 , such that

Conditional CDF

$$F_{X_1|X_2}(x_1, |x_2) = \lim_{\epsilon \rightarrow 0^+} \frac{P(X_1 \leq x_1, x_2 + \epsilon < X_2 < x_2 + \epsilon)}{P(x_2 + \epsilon < X_2 < x_2 + \epsilon)}$$

if it exists

and conditional PDF,

$$f_{X_1|X_2}(x_1|x_2) = \frac{f_{X_1, X_2}(x_1, x_2)}{f_{X_2}(x_2)}, f_2(x_2) > 0$$

$$F_{\underline{X_1|X_2}}(x|x_2) = \frac{\int_{-\infty}^x f(u_1, x_2) du_1}{f_{X_2}(x_2)}$$

$$\text{and, } F_{X_2|X_1}(x|x_1) = \frac{\int_{-\infty}^x f(x_1, u_2) du_2}{f_{X_1}(x_1)}$$

Normal Approximation formula:

$$P(S_n \leq x) \approx \Phi\left(\frac{x - E[S_n]}{\sqrt{\text{Var}(S_n)}}\right)$$

$$N(\mu, \sigma^2) = \sigma N(0, 1) + \mu.$$